Lovnesh Bhardwaj
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GitHub | Personal Site

Education

- **MSc — Artificial Intelligence (Magna Cum Laude)** *2026*
  Università della Svizzera Italiana, Lugano
- **BSc — Informatics (Magna Cum Laude)** *2023*
  Università della Svizzera Italiana, Lugano

Theses

- **Master's Thesis – Information Requirements and Allocation in Latent Representations:** Developed information-theoretic bounds connecting reconstruction quality, latent information requirements, and downstream task performance through rate-distortion theory. Investigated information allocation and disentanglement in VAEs, proposing a relevance-gated objective to encourage task-relevant information while suppressing nuisance information. [PDF]
- **Bachelor Thesis – Cryptocurrency Market Modelling Using Causal Inference Methods and Sentiment Analysis:** Developed a causal inference pipeline integrating VADER sentiment analysis with causal inference methods to evaluate the effect of social sentiment on crypto market movements, finding no significant causal effect of Twitter sentiment on crypto returns at standard significance levels. [PDF]

Research Interests

- Information Theory and Representation Learning
- Interpretable and Explainable Machine Learning
- Probabilistic Modelling and Uncertainty Quantification
- Time-Series Forecasting and Spatiotemporal Learning
- Causal Inference and Numerical Optimisation

Professional Experience

DXT Commodities — Developer *2022 – 2025*
Part Time

- Developed supply-and-demand curve forecasting models for day-ahead electricity pricing in Italy's NORD zone, combining historical auction curves with meteorological forecasts and energy-market fundamentals, building calendar-matched and fundamentals-adjusted forecasting baselines, and trained deep neural networks on bucketed curves with early stopping under monotonicity constraints.
- Developed a framework for Italian hydroelectric generation scheduling and water-value estimation, combining inflow models with constrained optimisation.
- Integrated Terna storage and generation data with ECMWF meteorological inputs; benchmarked seasonal and weather-based regression models against deep neural networks evaluated across multiple random seeds.
- Implemented linear programming and three-week rolling-horizon optimisation to estimate water shadow prices; compared model-implied valuations with hydroelectric offer-price distributions in the Italian day-ahead market (MGP), operated by GME.
- Developed a week-ahead forecasting model for hydroelectric supply offered at €0/MWh.
- Contributed to the development and training of transformer-based solar-radiation forecasting from satellite imagery; outperformed ECMWF forecast models for solar radiation forecasts up to 4 hours ahead on MAE and RMSE.
- Built WebSocket-based real-time data pipeline for high-frequency energy market data, enabling low-latency processing and analysis of electricity price fluctuations to filter out-of-distribution bids and offers in real-time using anomaly detection techniques.

Università della Svizzera Italiana — Teaching Assistant *2024 – 2025*
Part Time

- Taught and supported over 20 Bachelor students enrolled in Data Science, focusing on applied statistics and Data Analysis using Python and R.
- Designed programming exercises and tutorials to reinforce statistical modeling and data wrangling concepts.
- Led office hours and live sessions, helping students debug code, understand distributions, and interpret results from real-world datasets.
- Assisted instructors in grading and shaping the curriculum to enhance student engagement and learning outcomes.

Università della Svizzera Italiana — Computational Phylogenetics Research Intern *2022*
Summer Internship

- Improved maximum likelihood estimation in diversification models by analyzing numerical instability in Monte Carlo importance sampling for phylogenetic tree likelihoods.
- Optimized evolutionary model parameters using Differential Evolution on CSCS supercomputing cluster; identified scalability bottlenecks and proposed machine learning alternatives.
- Prototyped a neural network pipeline for parameter inference from branching times, trained on simulated phylogenetic trees, and evaluated performance against traditional likelihood-based methods.
- Designed preprocessing and data simulation tools for generating phylogenetic trees and training datasets under speciation/extinction constraints.

Conference Presentation

- **Innovation in dynamic network modelling — Bernoulli Society for Mathematical Statistics and Probability:** Modelled patent citation dynamics using Relational Event Models (REMs) with sampled partial likelihood, incorporating endogenous network effects alongside time-varying textual and technological similarity covariates; identified content similarity and preferential attachment as the dominant drivers of citation propensity. [Website] [Poster]

Manuscripts in Preparation

- **Relevance-Gated β-TCVAE:** Designs a training objective that decomposes the ELBO into relevance-aware components, applying sign-asymmetric per-dimension MI weighting to selectively reward information in relevant latent dimensions and suppress it in nuisance ones thereby constructing asymmetric behavioral incentives across a learned partition of the latent space — demonstrates competitive performance against state-of-the-art networks.

Ongoing Research

- **Mechanistic Interpretability of Transformer-Based Electricity Forecasts:** Investigates activation patching, attention-based analysis and sparse autoencoders to identify components contributing to forecasts at different horizons, with targeted ablations to assess their causal contribution.
- **Power-Market Structure in Time-Series Foundation Models:** Investigates whether model representations encode electricity-market fundamentals and merit-order relationships, and how these representations change following fine-tuning on market data.

Selected Projects

Selected code and project details available on GitHub or on request.

- **Spatiotemporal GNNs for Numerical Weather Prediction:** Designed and evaluated ST-GNNs to postprocess wind speed ensemble forecasts across 152 Swiss weather stations. Achieved 5% better CRPS score using a MultiScale Graph WaveNet, outperforming strong spatio-temporal baselines. [Code]
- **Swiss Insurance Chatbot:** Built a multilingual GenAI chatbot using retrieval-augmented generation (RAG), integrated with speech-to-text (Whisper) and text-to-speech (TTS) systems for full voice interaction. Tailored for regulatory Swiss insurance queries with modular evaluation and observability components. [Code]
- **ClimODE - Reproducibility Challenge:** Investigated the reproducibility of ClimODE, a physics-informed Neural ODE framework for climate and weather forecasting, by replicating its training and evaluation pipeline. Conducted GPU-based experiments and evaluated model checkpoints to assess the reproducibility of the original results. [Code]
- **Banking NLP:** Fine-tuned a transformer for supervised banking intent classification. Benchmarked classification accuracy and inference latency. [Available on Request]
- **Bug Prediction in Software Repositories:** Built a classifier to predict bug-prone files in software repositories using code quality metrics. [Code]
- **Multi-hop Influence Maximization:** Designed a GNN to approximate greedy influence (disease) spread in social networks using a custom fitness loss and GVAE-generated graphs. [Code]
- **CycleGAN Image Translation:** Trained a GAN-based model for unpaired style transfer; enforced cycle-consistency loss for domain alignment in visual mapping tasks. [Available on Request]
- **Amazon Recommender System:** Built SVD- and neural embedding-based recommenders; achieved RMSE of 1.05 (SVD) and 1.45 (deep NN) on Amazon Video ratings. [Code]
- **Quora Duplicate Detection:** Fine-tuned BERT for semantic duplicate detection, achieving 83% accuracy — 5 points above a TF-IDF + Logistic Regression baseline; studied the gap between surface-level n-gram similarity and deep semantic representations. [Code]
- **Maze Solver Robot:** Built a ROS-based autonomous robot using Thymio in CoppeliaSim; implemented flood-fill path planning, visual servoing with OpenCV, and PID-aligned motion control. [Code]

Technical Skills

- **Languages:** Python, Java, C/C++, MATLAB, R
- **Frameworks/Tools:** PyTorch, Scikit-learn, Pandas, NumPy, Linux, Git, Docker

Achievements

- **New York Academy of Sciences:** Invited to present a project on Food Fortification in New York, analyzing nutrient deficiencies and suggesting food solutions without requiring a blood test using regression techniques (2017).
- **Infosys Spark - Catch Them Young Program:** Selected among top students across Jaipur, India for a two-week residential training at Infosys Campus focused on programming, problem-solving, and system design (2015).